Optics

Questions with Fully Worked Answers · FormulaVerse PYQ Bank

Q1. For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle to minimum deviation will be _____.

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(1) 3:4
(2) 3:2
(3) 2:1
(4) 1:2

✓ Answer: (2) 3:2

Solution:

For a thin prism, $\delta_m = (n-1)A$, and at minimum deviation $i = (A+\delta_m)/2 = An/2$.
So $i/\delta_m = (An/2)/[(n-1)A] = n/[2(n-1)] = 1.5/(2\times0.5) = 1.5 = 3:2$.

Q2. Refer to the given figure. μ_1 and μ_2 are refractive indices of air and lens material. An object O of height 2 cm is at distance 40 cm from the refracting surface at pole P, with centre of curvature C at 20 cm from P. Given $\mu_1=1$, $\mu_2=1.54$. The height of the image will be _____ cm.

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(1) 1
(2) 0.5
(3) 1.2
(4) 0.25

✓ Answer: (1) 1 — unverified, see note below

Solution:

Using the single spherical refracting surface formula $\mu_2/v - \mu_1/u = (\mu_2-\mu_1)/R$ with $u = -40$ cm, $R = -20$ cm (based on the figure's left-to-right arrangement): solving gives $v \approx -29.6$ cm.
Magnification $m = (\mu_1 v)/(\mu_2 u) \approx 0.48$.
Image height = m × object height ≈ 0.48 × 2 ≈ 1 cm (nearest option).

Note: the exact sign convention depends on precisely which side C sits on in the original diagram — please verify this one against the figure directly.

Q3. In a single slit diffraction pattern, the wavelength of light used is 628 nm and the slit width is 0.2 mm. The angular width of the central maximum is $\alpha\times10^{-2}$ degrees. The value of α is _____.

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✓ Answer: 36

Solution:

Angular half-width to first minimum = $\lambda/a = (628\times10^{-9})/(2\times10^{-4}) = 3.14\times10^{-3}$ rad.
Full angular width of central maximum = $2\times3.14\times10^{-3} = 6.28\times10^{-3}$ rad.
Converting to degrees: $6.28\times10^{-3} \times (180/\pi) \approx 0.36° = 36\times10^{-2}$ degrees.
So $\alpha = 36$.
